How to read a backtest

A practical guide on real data · FX Trading Lab · kit v1.4 · generated 2026-11-06

Educational material. Not investment advice nor a recommendation to trade. Every number is a simulated result on historical data; past performance does not guarantee future results. The hypotheses shown here have demonstrated no edge; they are used precisely to teach how to check for one. Full notice: https://fx-trading-lab.pages.dev/legal/aviso

This guide uses two versions of the same intraday hypothesis on EUR/USD during 2024, V1 and V1.1, run with the same engine, the same data and the same costs. One lost. The other won a little. The question running through the whole document is whether that difference means anything, and how you would know.

What is in the kit

• 01 · This guide (ES and EN).
• 02 · Backtest report template, Markdown and PDF, with the sections we use in the lab.
• 03 · Decision checklist: the questions we go through before believing a number.
• 04 · Datasets: trades, equity curve, metrics, sensitivity and Monte Carlo for each run, with their SHA-256, plus the instructions to download the original M1 candles and reproduce everything.

How to use it

Read it once straight through (about 40 minutes). Then open the lab in your browser, pick a run and go through chapters 2 to 8 again with the screen in front of you: each chapter tells you exactly what to look at. To evaluate someone else's backtest, use the checklist in chapter 9; to document your own, use the template.

1. What an honest backtest is (and is not)

A backtest is a simulation: you take some rules, apply them to past data and write down what would have happened. It is cheap, fast and for that very reason dangerous: the computer gives you a number with two decimals even when the procedure is broken. An honest backtest meets four conditions, and if one fails, the number is worthless.

• **Causality.** At every instant the engine only knows what has already closed. A 09:00 M15 candle does not exist until 09:15:00; the decision is made then and execution happens at the next available price, never the same one.
• **Costs on every trade.** Spread (the provider's when available, otherwise a fixed one), slippage and commission. Without costs almost any intraday rule looks profitable.
• **Frozen hypothesis.** The rules live in a file with its hash. Changing a rule means creating another version (V1.1), not nudging V1 until the number looks nice.
• **Reproducibility.** Dataset hash, strategy version, execution configuration, seed, software version. If someone else cannot obtain the same number, it is not a result: it is an anecdote.

Both runs in this guide meet all four. That is why their numbers can be discussed. What follows is the discussion.

Field	Value (V1.1, EUR/USD 2024)
Dataset hash (SHA-256)	65a7a807004594d4a14e4408ce4abc71db1bc8f2d6c73e72a19f5ca25acde62d
Configuration hash	1ba09079be704d33d200276d5306ed197a93ced0b069adbc341c0eb62dccbbb3
Execution model	first_executable_m1_price_after_bos
Slippage	0.1 pips
Software	0.2.0 · source-sha256:e5bedf30a94f7bcaa37eaa733205a69121f4ac262b9649ff40e07f24f74fb81f

2. The five numbers that matter

A backtest summary usually comes with twenty metrics. Five are enough for the first reading, and it pays to always look at them in this order.

Metric	V1	V1.1	What it tells you
Trades (N)	5	12	How much evidence there is. Everything else is read in the light of N.
Expectancy (R per trade)	-0.52R	+0.20R	What it wins or loses, on average, each time it risks 1R.
Win rate	20 %	42 %	Only meaningful next to the payoff ratio.
Profit factor	0.24	1.54	Gross profit / gross loss. Below 1 it loses money.
Max drawdown (R)	3.45	3.01	The worst run, measured in risk units, not in euros.

Why everything is in R

R is the distance to the stop on each trade. Winning +1.8R means you made 1.8 times what you risked on that trade; losing -1R is losing exactly the planned risk. Measuring in R makes trades with different stops comparable and takes account size and leverage, which are decisions at another level, out of the way. When a backtest shows you annual return percentages, ask for the R: a +40 % may be +4R at 10 % risk per trade, i.e. an account that would have blown up in the first bad streak.

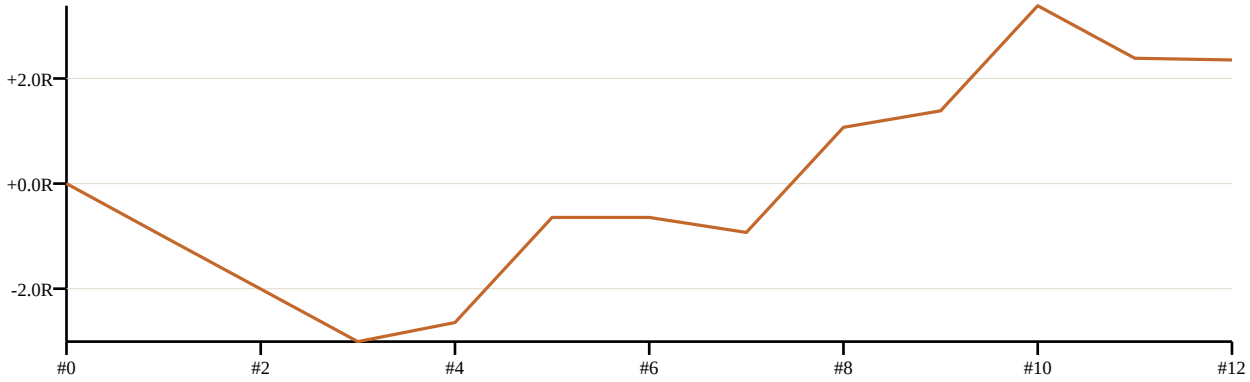
Win rate and payoff go together

V1.1 wins 42 % of the time. That sounds bad until you see the average win is +1.34R and the average loss -0.62R: a payoff ratio of 2.16. With that ratio, winning 32 % is enough to break even. The average loss is smaller than 1R because 5 of the 12 trades were closed at session end, not at the stop: the engine never holds a trade overnight, which cuts losses but also cuts gains.

Exit reason	Trades
session expiry	5
stop loss	4
take profit	3

3. The equity curve: what to look at

V1.1 · cumulative R per trade (EUR/USD 2024)



The equity curve is the most looked-at metric and the worst read. Three things before drawing conclusions:

• **The X axis is trades, not time.** Twelve steps across twelve months. On a date axis you would see eleven months of flat line and a couple of jumps. Both views are true; the second is more honest about how long you spend doing nothing.
• **Where the drawdown is.** The maximum (3.01R) comes with 3 consecutive losses. With N = 12 that is not a rare streak: it is the norm, as chapter 4 shows.
• **How many trades explain the result.** Remove the best trade and add up again. If the total changes sign, the result depends on one particular day.

Let us do the last one with V1.1: the best trade was +2.00R on 2024-04-25. Without it the total goes from +2.35R to +0.35R. A single trade flips the sign of the year. Write that down: it is the most important fact in the whole report, and it appears in no metrics table.

4. Twelve trades prove nothing: Monte Carlo

If you shuffle the order of V1.1's twelve trades you get the same total but different drawdowns. If instead of shuffling you resample (draw twelve trades at random from the twelve, with replacement) you get different totals. Do it a thousand times and you have a distribution of what the same strategy could have produced with the same luck and another sequence. That is a Monte Carlo, and this table is how to read it.

Percentile	Final R · V1	Final R · V1.1	Max drawdown · V1.1
P5	-5.00R	-3.65R	1.01R
P25	-3.90R	-0.34R	2.00R
P50	-2.61R	+2.20R	2.39R
P75	-1.32R	+4.97R	3.58R
P95	+0.52R	+8.71R	5.29R
Observed	-2.61R	+2.35R	3.01R
Probability of ending negative	94 %	28 %	

Read the V1.1 column: between P5 and P95 the outcome runs from -3.65R to +8.71R. That is the width of what you do not know. A strategy that made +2.35R in 2024 had a 28 % chance of ending the year negative with the same trades in another order or a similar sample. V1 is the opposite case: 94 % probability of losing. There the Monte Carlo does not tell you V1 is bad; it tells you that, with five trades, almost anything is compatible with what was observed.

Rule of thumb. If P25 is below zero, the backtest has not shown anything yet, whatever the observed case made. And the width between P5 and P95 only narrows with more trades: not with more metrics, not with more charts.

5. Would chance do the same? The random benchmark

The most uncomfortable question: what would have happened entering at random inside the same time windows, with the same stop, the same target, the same costs and the same session-end exit? The lab simulates it 200 times. If the strategy does not clearly beat those random entries, what you are looking at is the market's behaviour at that hour, not the rule.

	V1	V1.1
Strategy expectancy	-0.52R	+0.20R
Random expectancy · P50	-0.10R	-0.07R
Random expectancy · P95	+0.55R	+0.54R
Strategy percentile vs random	8 %	76 %
Empirical p-value (one-sided)	0.92	0.24

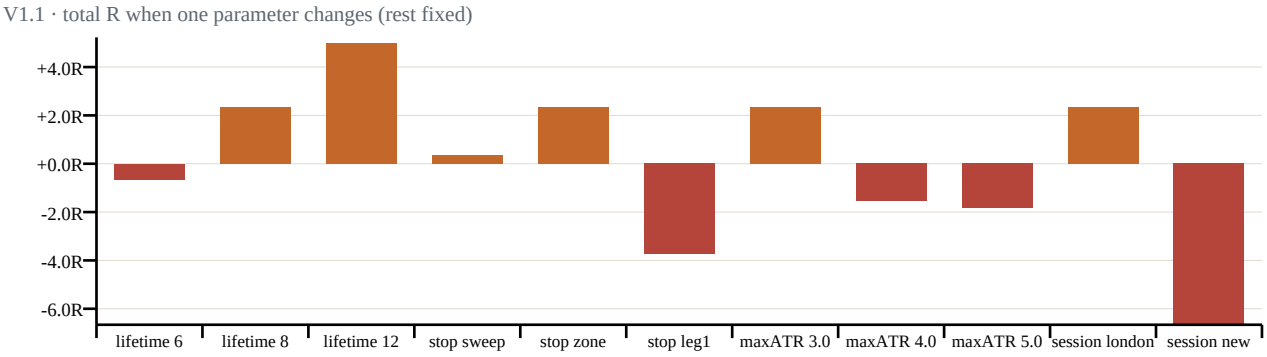
V1.1 sits at the 76th percentile of random entries: better than three out of four, worse than one in four. A p-value of 0.24 means one in four strategies entering at random would have done just as well. Nobody would publish that as evidence in any serious field. The usual bar ($p < 0.05$) would require beating 95 % of random entries, and even then with $N = 12$ it would need repeating in another year.

What you do learn here. The median random expectancy is negative in both versions: entering at random at the London open with those costs loses money. That is the baseline any rule must be measured against; without it, +0.19R looks like a lot and you do not know relative to what.

6. Fragility: the sensitivity test

A robust rule looks like itself when you move a parameter a little. A fragile rule flips sign. The lab re-runs the hypothesis changing one parameter at a time and keeping the rest fixed. This is V1.1:

Parameter	Value	N	Total R	Expectancy
target.baseline	1.5R	12	+2.35R	+0.20R
target.baseline	2.0R	12	+2.35R	+0.20R
target.baseline	2.5R	12	+3.85R	+0.32R
target.baseline	3.0R	12	+4.06R	+0.34R
slippage.baseline_pips	0.0	12	+2.46R	+0.21R
slippage.baseline_pips	0.1	12	+2.35R	+0.20R
slippage.baseline_pips	0.2	12	+2.26R	+0.19R
slippage.baseline_pips	0.5	11	+2.06R	+0.19R
risk.baseline_equity_risk_fraction	0.0025	12	+2.35R	+0.20R
risk.baseline_equity_risk_fraction	0.005	12	+2.35R	+0.20R
risk.baseline_equity_risk_fraction	0.01	12	+2.35R	+0.20R
zones.expiry_m15_bars	4	12	+2.35R	+0.20R
zones.expiry_m15_bars	8	12	+2.35R	+0.20R
zones.expiry_m15_bars	12	12	+2.35R	+0.20R
setup.lifetime_m15_bars_from_sweep_confirmation	6	9	-0.65R	-0.07R
setup.lifetime_m15_bars_from_sweep_confirmation	8	12	+2.35R	+0.20R
setup.lifetime_m15_bars_from_sweep_confirmation	12	14	+4.98R	+0.36R
displacement.minimum_atr	0.75	12	+2.26R	+0.19R
displacement.minimum_atr	1.0	12	+2.35R	+0.20R
displacement.minimum_atr	1.25	11	+3.35R	+0.30R
stop.reference	sweep_extreme	13	+0.35R	+0.03R
stop.reference	zone_distal	12	+2.35R	+0.20R
stop.reference	leg1_origin	18	-3.74R	-0.21R
stop.max_distance_atr	3.0	12	+2.35R	+0.20R
stop.max_distance_atr	4.0	17	-1.55R	-0.09R
stop.max_distance_atr	5.0	19	-1.82R	-0.10R
liquidity.level_max_age_m15_bars	None	15	+3.29R	+0.22R
liquidity.level_max_age_m15_bars	960	10	+0.04R	+0.00R
liquidity.level_max_age_m15_bars	1920	12	+2.35R	+0.20R
liquidity.level_max_age_m15_bars	3840	12	+2.35R	+0.20R
session	london_open_09:00-10:30	12	+2.35R	+0.20R
session	new_york_open_15:00-16:30	8	-6.66R	-0.83R



How to read the table: look for the rows where the result flips sign or multiplies.

• **stop.reference.** With the stop at the sweep extreme or at the impulse origin, V1.1 makes almost nothing or loses. The positive result depends on a single stop choice. It is the most important finding of 2024 and the reason V1.1 exists (V1 used another stop).
• **setup.lifetime.** Six candles of lifetime instead of eight: negative. Twelve: double. A parameter that comes from no theory decides the sign.
• **session.** The same rule at the New York open clearly loses. It is not a strategy: it is, at most, an observation about the London open.
• **What changes nothing** (target, risk per trade, zone expiry) is not good news either: it means those parameters almost never come into play with twelve trades.

Rule of thumb: if moving a parameter one step flips the sign of the total, log that parameter in the decision log as *unresolved* and do not optimise on it. Optimising a fragile parameter on twelve trades is fitting noise.

7. Out of sample: splits and walk-forward

Everything above is computed on the full year, which is also the year the rule was designed on. The only way to know whether a rule holds on data it never saw is to set aside a stretch and not touch it until the end. The lab does two things: a chronological split (train, validation and out-of-sample) and a walk-forward with rolling windows.

Segment (V1.1)	Days	N	Total R	Expectancy
train	158	8	+1.07R	+0.13R
validation	53	3	+1.31R	+0.44R
out of sample	52	1	-0.03R	-0.03R

Walk-forward (concatenated out-of-sample)	V1	V1.1
Trades	3	6
Total R	-0.66R	+3.00R
Windows with trades / positive	3 / 1	4 / 2
Consistency	33 %	50 %

V1.1's out-of-sample segment has 1 trade. One. Nothing can be concluded, and that is exactly what must be written in the report. The walk-forward yields 6 out-of-sample trades with 2 positive windows out of 4: a coin toss. An intraday strategy with twelve trades a year needs several years of data just for the out-of-sample split to make sense. That is why the lab's next step is 2022-2023, not another metric.

8. Costs: from positive to negative

Slippage (pips)	N	Total R	Expectancy
0.0	12	+2.46R	+0.21R
0.1	12	+2.35R	+0.20R
0.2	12	+2.26R	+0.19R
0.5	11	+2.06R	+0.19R

Here V1.1 behaves well: going from 0 to 0.5 pips of slippage takes about 0.4R off twelve trades, because the stop is far (several ATR) and costs are small relative to risk. In strategies with stops of a few pips the same table usually flips sign on the second row. Always ask for this table; if it does not exist, assume the worst case.

Three cost questions you must be able to answer by reading the report: where does the spread come from (provider history or a fixed number)? Is slippage applied to the stop as well? Is commission in the same unit as the risk?

9. The decision checklist

Twelve questions. If any of the first five fails, stop reading the report. If two of the rest fail, the verdict is *we do not know yet*, which is the verdict of almost every honest backtest.

- 1. Are the dataset hash, strategy version and execution configuration published?
- 2. Does the engine decide only on closed candles and execute at the next price?
- 3. Are spread, slippage and commission applied to every trade, and is their source stated?
- 4. Were the rules written before seeing the result (hashed file, named version)?
- 5. Can the number be reproduced with the code and data indicated?
- 6. How many trades are there? What happens to the total if you remove the best one?
- 7. Monte Carlo: is P25 above zero? What is the probability of ending negative?
- 8. Random benchmark: does the strategy beat 95 % of random entries with the same costs?
- 9. Sensitivity: does any parameter flip the sign of the total when moved one step?
- 10. Out of sample: is there a reserved segment with enough trades to say anything?
- 11. Costs: does the result survive 0.5 pips more slippage?
- 12. Does the report end with a research decision (discard, continue, new version) and not with an invitation to trade?

The checklist is also a separate file (03-checklist) to print or paste at the end of every report.

10. How to tell a result (especially a negative one)

A backtest report is not a brochure. Its job is to let someone else redo it and disagree with data. The kit's template (02-plantilla) has these sections, in this order, and each has a reason:

• **Hypothesis and version.** Which rule, in which file, with which hash. What changed from the previous version and why (link to the decision log).
• **Data.** Provider, symbol, range, timezone, hash, gaps and invalid rows.
• **Execution and costs.** Fill model, spread, slippage, commission, risk per trade.
• **Result.** The five numbers, the R curve, the exit reasons. And the total without the best trade.
• **Protocol.** Monte Carlo, random benchmark, sensitivity, splits and walk-forward, each with its table.
• **Limitations.** The ones you already knew and the ones the protocol taught you.
• **Decision.** One of three: discard, keep researching (with which new data), or move to the next version (with which exact change). Never *trade it*: that is not a research decision.

A well-documented negative result is worth more than an undocumented positive one: it saves you repeating the mistake and saves others believing the number. V1 lost; from its report came the stop finding and, from there, V1.1. That chain, not the +2.35R, is the lab's product.

11. Glossary

Term	Meaning
R	Risk unit: the distance to the stop on each trade. +2R = twice the risk taken.
Esperanza	Average R per trade. Positive does not mean profitable when N is small.
Factor de beneficio	Sum of gains / sum of losses (absolute value).
Drawdown	Fall from the previous peak of the equity curve, here in R.
MFE / MAE	Maximum favourable / adverse excursion of a trade before it closes, in R.
Monte Carlo	Resampling of the trades to see the distribution of possible outcomes.
Benchmark aleatorio	Random entries with the same exit rules and costs; the baseline the rule is measured against.
Sensibilidad	Re-run changing one parameter at a time.
Walk-forward	Rolling windows: fit on one, evaluate on the next, without overlap.
Fuera de muestra	Reserved stretch of data not used to design or tune the rule.
Look-ahead	Using information that did not yet exist at decision time. Invalidates the backtest.
Hash (SHA-256)	File fingerprint. If one byte changes, the hash changes: it lets you know two people used the same data.

12. Reproducing these numbers

All the code is public. The kit's datasets (04-datasets) contain the trades, the curve, the metrics, the sensitivity and the Monte Carlo of each run, with the SHA-256 of every file. The original M1 candles are not included because they belong to the provider; the file COMO-DESCARGAR-DATOS.md explains how to download them with the repository script and check that their hash matches the one in this guide.

Resource	URL
Lab (free)	https://fx-trading-lab.pages.dev
Code	https://github.com/inigogonzalezgarcia/9999-trading-lab
Legal notice and method	https://fx-trading-lab.pages.dev/legal/aviso

13. Epilogue: what happened in 2022 and 2023

The previous chapters were written with 2024 and ended with a promise: the answer was in more years, not more metrics. Here is the answer. Same V1.1 rule, same engine, same costs (fixed 0.3-pip spread because the 2022-2023 data has no spread column; the real 2024 median was 0.2).

Year	N	Total R	Expectancy	Win rate	Percentile vs random	P(losing) (MC)
2019	5	-0.84R	-0.17R	40 %	45 %	63 %
2020	15	-0.10R	-0.01R	40 %	51 %	52 %
2021	7	+3.86R	+0.55R	71 %	94 %	11 %
2022	11	-8.01R	-0.73R	18 %	1 %	100 %
2023	4	-0.29R	-0.07R	25 %	52 %	62 %
2024	12	+2.35R	+0.20R	42 %	76 %	28 %
2025	10	-0.96R	-0.10R	50 %	46 %	62 %
2026	3	+1.94R	+0.65R	67 %	86 %	14 %
8 years	67	-2.04R	-0.03R			

2022 is the full answer: 11 trades, -8.01R, 9 of them closed at the stop, worse than 99 % of random entries with the same exits and costs. No parameter in the sensitivity table rescues it: every stop reference, setup lifetime and max-distance variant stays between -6.8R and -10.9R. The "stop finding" of chapter 6 was a finding about 2024, not about the rule.

Go back to chapter 4 and look at the 2024 column: P25 below zero, 28 % probability of losing, 75th percentile against chance. The protocol already said "we do not know yet". What 2022 and 2023 do is turn that "we do not know" into "no". That is the normal sequence for an honest hypothesis, and it is why the 2024 report did not say "trade it".

• **Logged decision.** V1.1 is discarded as an improvement over V1 and stays published as a worked example of a false positive. No V1.2 is derived from 2024.
• **New rule for the lab.** No hypothesis is read on a single year: first the three years pooled, then each year on its own to see the dispersion.
• **For your own backtest.** If you only have one year, what you have is a chapter 4. Ask for the second year before believing anything, and expect it to be worse.

Educational material. Not investment advice. Past performance does not guarantee future results.